

**ATHENS UNIVERSITY OF ECONOMICS AND BUSINESS
DEPARTMENT OF INFORMATICS**

**Wireless Networks and Mobile Communications
Fall Semester 2025-26**

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**Assignment #1 – 14 October 2025
Due date: 7 November 2025**

The aim of the first experimental assignment is to get familiar with channels in Wi-Fi local area networks (IEEE 802.11), signal attenuation, interference from neighboring networks, and link (connection) speed.

Devices that you will need: i) a laptop or desktop computer with Wi-Fi connectivity, ii) a smartphone, and iii) a wireless Access Point, which can be within the DSL model of your broadband connection (referred to below as “AP”). Instead of a smartphone you can use a second laptop or desktop computer. When conducting the measurements you are advised to keep your smartphone and laptop on a stable surface.

Software that you will need: A Wi-Fi analyzer mobile application that reports the wireless networks within its range and the corresponding received signal strength and ii) a similar application for a laptop/desktop computer.

Experiments

a) Select three positions in your house, the first close to the AP, the third far from the AP, and the third somewhere in between the first and third positions. With the Wi-Fi analyzer identify for each of the three positions (independent of the Wi-Fi channel they are transmitting) the number of wireless networks (the neighboring wireless networks and your wireless network) within the range of your smartphone that have signal strength (dBm) 1) higher than -60dBm, 2) in the interval -70dBm to -60dBm, 3) smaller than -70dBm. The signal strength may vary with time. For this reason, you can count the number of neighboring wireless networks at three different time instances, with a difference of 2 minutes before each instance. Present your measurements in a table that has a different line for each of the three positions where you performed the measurements and a different column for the three measurements.

b) Select two additional positions in your house, between the first and second position and the second and third position, respectively, that you initially selected for the previous question. For all five positions measure the signal strength and the Wi-Fi link (or connection, network) speed and present your results in a table that has a different link for each of the five positions in your house. The speed of your Wi-Fi interface can be found through the connection information (on a smartphone) or the Wi-Fi status on your laptop/desktop computer. In the table, add a column with an estimation of the distance between the measurement position and the AP and a column with the number of walls between the measurement position and the AP.

Using the measurements of this question, create a graph showing the speed of the Wi-Fi interface as a function of the signal strength. Do the above, i.e., create a table with the signal strength measurements and connection speed and a graph, for both the smartphone and laptop/desktop. What can you say concerning the dependence of the connection speed on the signal strength? Are there differences between the measurements on the smartphone and those on the laptop/desktop? If the answer is yes, what can be the cause for these differences?

c) For the closest and the farthest position you selected in question a), and for the wireless networks you reported in this question with signal strength above -60dBm and with signal strength between -70dBm and -60dBm, assign the name A1, A2, etc, for each wireless network within the range of your smartphone and identify the channel it is transmitting. Present the measurements in a table, one for the closest position and one for the farthest position, which should include the signal strength and channel of all wireless networks within the smartphones range, along with those of your wireless network. The lines in your table should be grouped in channels in the 2.4GHz band and in the 5GHz band. Based on the information in the two tables (one for the closest position and one for the farthest), which channel would you suggest for your own network? Justify your answer. How does your suggestion compare with the suggestion (or rating) from the Wi-Fi analyzer application?

Note:

Your wireless network and your neighbors' wireless network may be configured to transmit in both 2.4GHz and 5GHz bands (dual band). In this case, for question c) the signal strength for both bands should be included in the tables.